Steam-LLMSonas: Simulating Community Reaction to Monetization Decisions

Abstract

Studios that own a live, premium game face a recurring decision: development cycles are long and release-day outcomes are volatile, so many firms add recurrent monetization (microtransactions, battle passes, cosmetic shops) to stabilize revenue between releases. That decision has a predictable cost in community goodwill, but the cost is usually estimated after the fact, from the review-bomb itself. This submission asks whether an LLM-persona panel, built from real players' pre-event behavioural histories and never shown the outcome, can predict that cost in advance. I built ∼100 personas per case from real Steam reviewers' histories, ran them through a ladder of construction and readout methods, and scored every method against Steam's own recommend-ratio history around dated monetization events, so every number reported is a held-out prediction, never a fit. The submission includes the scored ladder across a set of pillar and umbrella cases plus one scored control case, counterfactual decision menus that rank the levers a studio actually controls, a second behavioural question testing whether the same personas predict what players do as well as what they say, and a set of diagnostic ablations that stress-test the method's honesty. Across the five scored monetization cases, the grounded panel's predicted severity ranking tracks the real one, with two essentially exact hits. One of them is on the single case whose event postdates the model's training data, ruling out memorization as an explanation. The method's failure modes (a case-specific normative level, and a hard boundary between predicting sentiment and predicting behaviour) are measured directly and reported as central results, not smoothed over.

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1. The business problem & stakeholder question

The games industry's development cycles are long and its release outcomes are volatile: years of work can land on poor sales, and that structural risk is part of why the industry's outlook is under pressure right now. A common hedge is recurrent monetization: one product that can be monetized continuously, microtransactions being the classic form, funds the riskier release pipeline around it. That is the position this submission's stakeholder is in.

A publisher who already sells a premium (paid) game is considering adding microtransactions, or, for a game that already has some recurrent monetization, expanding it further (a new shop, a battle pass, a subscription tier). The commercial logic is straightforward: development cycles for a new title or a major expansion can run for years, and a single release's commercial performance is hard to predict, so recurrent monetization inside an existing live game is an attractive way to smooth revenue between big bets. The question the stakeholder actually asks is not abstract. It is "what would happen to our community if we did this, and is there a way to do it that costs us less goodwill?"

That question has two parts a simulator needs to answer honestly. First, the direction and magnitude of the reaction: will the community's approval fall, and by roughly how much? Second, the shape of the decision: given that some version of the monetization change is going to happen, which implementation choices (disclosure, pricing model, whether the added content affects gameplay balance) matter most to the size of the backlash? A studio cannot act on "players will be upset"; it can act on "disclosing the shop upfront recovers most of the goodwill that removing it entirely would."

This submission treats both parts as things a validated persona panel should be able to answer, and keeps the two carefully apart: the scored ladder (§4, §5.1–5.3) answers the first question against real observed outcomes; the counterfactual decision menus (§5.4) answer the second, and are explicitly labelled as extrapolation beyond any ground truth, anchored only where the shipped decision matches a real case.

2. Why Steam + why retrospective ground truth

The central design decision in this project is methodological, not domain colour: only ask a question whose real answer can be observed, so every model score is a genuine prediction rather than an unfalsifiable narrative. Video games satisfy this unusually well. Steam exposes a public, timestamped review histogram for every title (an approval-through-time curve that anyone can pull), and Steam's news feed (`ISteamNews`) timestamps the events (patches, announcements, DLC releases) that move it. That combination means a monetization change can be dated precisely, and the real before/after split in player recommend rate around that date can be read directly from Steam's own data, rather than inferred, surveyed after the fact, or assumed.

This gives the project a genuine circularity guard: personas are constructed exclusively from each reviewer's *pre-event* behavioural history (library composition, spend pattern, review timing, tenure) and are then scored against *post-event* sentiment they were never shown. Because the event is already in the past, the ground truth exists before the model ever answers, but the model's construction inputs are cut off before the event, so the score is a held-out prediction, not a fit to known data. I've stated the honest caveat that comes with this design up front rather than glossed over it: Steam reviewers are vocal, engaged players, not a random sample of the median purchaser. But because the ground truth is drawn from that same reviewing population, the persona panel and the population it is scored against are aligned by construction, which is the property that matters for the comparison to be meaningful.

3. The cases & the question

Every persona in every case is asked the same fixed, neutral, third-person question:

> "Would this player recommend the game to other players after this change?"

The question itself carries no valence and names no decision: the decision being evaluated lives entirely in the persona's situation clause (the `change` field described in §4.1), not in the question's wording. Why the question is built this way, and what was measured when it was built differently, is covered in §4.5.

Cases are organized into three tiers by role: pillar cases sit squarely in scope (a premium game adding microtransactions for the first time), umbrella cases extend the claim to a related but distinct decision (a game that already has some recurrent monetization expanding it further), and controls are cases deliberately chosen *not* to be premium-adds-MTX events, used to show the method responds to real content rather than firing on any dated patch.

Case	Event	Date	Decision type	Role
Payday 2	Addition of purchasable, pay-to-win-adjacent weapon/skin microtransactions, breaking a prior public no-MTX promise	2015	Premium game adds MTX (flagship)	Pillar

Case	Event	Date	Decision type	Role
Tekken 8	Launch of the “Tekken Shop” real-money cosmetics store, followed by a paid battle pass	2024	Premium game adds MTX	Pillar
Elite Dangerous	Addition of a cosmetics/MTX store to a premium space-sim	2019	Premium game adds MTX (low-reaction case)	Pillar (the calm case)
War Thunder	Economy and monetization revolt over expanding existing recurrent-monetization systems	2023	Expansion of recurrent monetization in an already-monetized game	Umbrella
RuneScape	Introduction of the “Hero Pass” battle-pass-style system	2023	Expansion of recurrent monetization in an already-monetized game	Umbrella
CS:GO	Transition from paid to free-to-play	2018	Price removal, not a monetization addition	Control
No Man’s Sky	Major free content update (NEXT)	2018	Free content update, not monetization	Control
Helldivers 2	Account-linking (PSN) backlash	2024	Governance and integrity issue, not monetization	Control (GT-integrity example)

The pillar cases carry the headline claim. The umbrella cases test whether the same construction and scoring machinery generalizes to a related but distinct studio decision. The controls exist to show that the method’s predictions track the actual content of an event rather than a generic “players get upset at patches” prior: each control was chosen because its correct behaviour is a *different* shape of reaction (a price cut should not read as a betrayal; a free content update should not read as a backlash; an account-linking dispute is a governance complaint, not a monetization complaint). Of the three, only No Man’s Sky is carried through to a scored result in §5.1; CS:GO and Helldivers 2 informed case selection and are retained here for that reason but do not appear as scored numbers.

Two further internal distinctions matter and are named explicitly. Within the pillar tier, Payday 2 and Tekken 8 are the two genuine “premium game quietly adds MTX later, on a clean pre-event window” specimens identified; Elite Dangerous is retained as a contrast within the same decision type precisely because its community reaction was comparatively muted, which tests whether the method can also predict a *small* effect rather than always predicting a large one. Second, Tekken 8’s event postdates the scoring model’s training cutoff, which removes any possibility that a strong result there reflects memorized outcome data rather than genuine prediction from behavioural features.

4. Method

4.1 Persona construction

Each persona is built from one real Steam reviewer's pre-event behavioural history, never from their review text, which is deliberately excluded from the bio because it is the withheld verdict the model is being scored against. Construction proceeds in four parts:

Population-relative segmentation. Every behavioural marker is expressed relative to the case's own population, not as an absolute threshold, using quantile bins computed mechanically and frozen before any model is run or any ground truth is consulted. This keeps segmentation identical in procedure across cases, so a "heavy spender" or "vocal reviewer" label means the same thing (top quantile within that case's population) everywhere, while allowing the underlying scale to differ case to case.

Facts-only bios. The persona's biography is composed entirely of verifiable facts about the account's pre-event history: library breadth, investment level, purchase channel, review timing and tenure, vocalness (review frequency/length as a behavioural signal, not content), and an early-access flag where relevant. The bio is written in the third person and contains no opinion, sentiment, or forecast: the model is told what this person did, never what this person thinks. A real constructed bio, from the Payday 2 run, illustrates the form:

> This Steam player is a casual player of the game, with about 54 hours in it. They almost never write reviews (2 to date). They paid for the game on Steam. They have owned and played it since long before this, a long-time owner. The developer had said the game would never have microtransactions, and has now added purchasable drills that open safes containing weapon skins, some of which carry stat bonuses.

Monetization-exposure and disposition markers. Two additional axes are computed and included in the bio: an exposure marker (how much this account's pre-event spending pattern already involved optional purchases, e.g. a "free-to-play-experienced" vs "premium-only" split) and a disposition marker derived from this reviewer's general historical recommend rate across their library (a coarse "tends to recommend / mixed / tends to criticize" signal, computed only from pre-event data). These are the two axes the ladder result is checked against for internal consistency: a valid method should order its answers along disposition and exposure even when the aggregate level is off.

Guardrails. Bins are mechanical and frozen before scoring; review text is excluded; the survey question itself is written to be neutral, so that the decision being evaluated lives in the persona's bio, not in leading language in the question. These guardrails are applied identically across every case, not tuned per case.

4.2 The method ladder

Four methods are compared, forming a bake-off scored against the same ground truth in every case:

- **M1: Naive prompting (control).** A minimal "act as a gamer" persona with no grounding in real behavioural data, asked the same question in isolation. This is expected to plateau near the model's generic prior and serves as the floor the grounded methods must clear.
- **M2a: Real sampled users.** A stratified sample of ~100 real reviewers' pre-event histories, each replayed as one persona.
- **M2b: Cluster archetypes.** ~100 behavioural cluster archetypes, with coverage forced from the data (anti-collapse) so that the persona set reflects the real population's diversity rather than collapsing onto its mode.

- **M3: Grounded + social graph.** The stronger of M2a/M2b placed on a homophily-weighted graph and updated via Friedkin–Johnsen (anchored) dynamics before answering, testing whether modelling social influence between personas improves on independent answers, expected to help most where the underlying population is genuinely divided rather than uniform. (One caveat here, our chosen data makes these graphs difficult to establish, we do a shuffle null validation on these but to be honest I dont know if I figured out the best way to implement these)

4.3 Survey execution

Each persona is asked a single-select question about the case's decision, written in the third person and kept neutral ("which option does this player pick?" rather than a framing that invites a socially desirable answer): the one mitigation the social-desirability-bias literature found reliable for this failure mode. Rather than free-generating an answer and sampling repeatedly, the survey reads the model's option-token logprobs directly at temperature-irrelevant precision, which is both cheaper and avoids conflating sampling noise with genuine belief.

A further readout refinement, adopted onto the scored path during development, is **permutation-averaged readout**: every persona is surveyed twice, once with each answer option listed first, and the two resulting probabilities are averaged. This exists because option-position (not option identity) was found to bias the model's answer by a material margin; averaging over answer order is mechanical, decided before any ground truth is examined, and case-blind, so it removes a real source of systematic error without giving the model any information about the case it didn't already have. It is applied uniformly to every method, including the M1 control, so comparisons across the ladder remain fair.

4.4 Scoring

The primary score is Jensen–Shannon divergence between each method's predicted answer distribution and the real recommend-ratio distribution observed on Steam around the event. Because Steam enforces one review per user, the pre-event reviewer population and the post-event reviewer population are structurally disjoint sets of accounts: a person who reviewed before the event and never edits that review does not appear in the post-event data, and a post-event reviewer may never have reviewed before. This means there is no single population that is simultaneously "the people the personas were built from" and "the people the post-event recommend rate was measured from."

The honest target is therefore not a single number but a **bracket**, bounded by two legitimate populations: the **panel** (existing pre-event reviewers who edited their review inside the post-event window: the population closest to what the personas actually represent) and the **flow** (all reviews newly posted inside the post-event window, a different, partially-overlapping population). A method's score is reported against both ends of this bracket, and a result is judged by where it falls relative to the bracket, not against a single point estimate: reporting only one end would overstate the precision the data actually supports.

4.5 Design defenses

Five choices in this method were contested during development. Each is stated here with the evidence that settled it.

Neutral third-person question wording. The survey question (§3) names no decision and carries no valence; the decision lives in the persona's situation clause, not the question. This is the one social-desirability mitigation the SDB literature found reliable: third-person reformulation, per *Mitigating Social Desirability Bias in Random Silicon Sampling* (Chapala, Mironov et al.), which also found that "sincerity" or analytic meta-instructions backfire by raising uniformity rather than reducing bias.

Logprob readout over sampled text. The survey reads option-token logprobs directly rather than free-generating an answer and sampling repeatedly, both because it is cheaper (one forward pass per persona per label order) and because temperature-based sampling does not reliably recover a model's internal belief split. The verbalized alternative was tested directly and failed: asking the model to state a rate out of 100 rather than reading the logprob gives 0.535 against Payday 2's bracket [0.113, 0.305], an overshoot from above, with 99 of 100 answers landing on multiples of 5: heavy round-number clumping rather than a genuine distribution (§5.6, E2).

Permutation-averaged readout. Before this fix, option *position* (not identity) biased the model's answer by a median of about 2.4 nats toward whichever option was listed second (E1, §5.6). The fix, surveying every persona under both label orders and averaging the two probabilities, is mechanical, case-blind, and was adopted before any ground truth was consulted for it, and it is applied uniformly to every method including the M1 control. This label-position bias and a permutation-based fix are also reported independently in *Large Language Models Are Not Robust Multiple Choice Selectors* (Zheng, C., et al., arXiv:2309.03882), which shows LLM MCQ answers are biased by option position via token prior and proposes PriDe, a label-free inference-time debiasing method that estimates the option-ID prior by permuting option contents: published corroboration of both the bias this project measured and the permutation-averaging direction of the fix.

The bracket estimand. Already established in §4.4: Steam's one-review- per-user rule makes the pre-event and post-event reviewer populations structurally disjoint, so the honest target is a [panel, flow] bracket rather than a single number.

Model choice. The scored path uses Llama-3.3-70B-Instruct-Turbo, chosen in part because it exposes open logprobs. Claude was ruled out on this basis alone, since the Anthropic API does not expose token logprobs. Because the model's own case-specific normative read is the source of the method's sharpest failure mode (§5.6), it matters whether that failure is a Llama-specific quirk. It is not: the E3 cross-model probe shows the same normative-confidence collapse on DeepSeek-chat is more severe, not less: the Recommend token falls below DeepSeek's top-20 logprob reporting floor for all 20 of 20 personas tested, a deeper collapse than anything observed on Llama, which at least keeps the option token within finite reporting depth. The model choice does not drive the findings reported here.

5. Results

All results below use a single fixed run configuration, stated once here rather than repeated per case: Llama-3.3-70B-Instruct-Turbo, n=100 personas per case, seed 42, drawn from a pool of 3,000 pre-event English-language reviewers, scored under the permutation-averaged binary logprob readout (§4.3), against the [panel, flow] bracket (§4.4). The post-event ground-truth window is 7 days for every case except Tekken 8, which uses 45 days, a window chosen from the shape of the swing itself, before any model was run on the case, because that case's backlash is a slow burn rather than a launch-day spike (§5.2).

5.1 Headline methods × cases table

The scored ladder answers one broader question in two tiers. The three pillar cases (Payday 2, Tekken 8, and Elite Dangerous, all premium games that added microtransactions after release) answer it in the narrowest form a stakeholder would actually ask: "how will my community respond if I add microtransactions to my already-released premium game?" The two umbrella cases (War Thunder and RuneScape, both already-monetized games that attempted to increase recurrent monetization further) extend the same question to its natural generalization: "how would my community respond if I attempted to increase monetization via microtransactions?", whether the starting point is a clean premium title or a game that already has some shop or subscription in place. The five cases together are the evidence base for that broader question, and the table below is organized by tier so the two questions stay legible as distinct rows rather than one undifferentiated list.

Pillar tier: premium game adds microtransactions

Case	Prior	Bracket [panel, flow]	M1	M2a	M2b	M3
Payday 2	0.913	[0.113, 0.305]	0.975 (JSD 0.4150)	**0.064** [0.034, 0.095] (JSD 0.0743)	0.063 (JSD 0.0755)	0.049 (JSD 0.0888)
Tekken 8 (held out)	0.835	[0.261, 0.560]	0.973 (JSD 0.2004)	**0.526** [0.435, 0.619] (JSD 0.0009)	0.571 (JSD 0.0001)	0.441 (JSD 0.0103)
Elite Dangerous	0.721	[0.616 flow, 0.667 panel]	0.977 (JSD 0.1687)	**0.917** [0.858, 0.967] (JSD 0.0972)	0.917 (JSD 0.0971)	0.909 (JSD 0.0903)

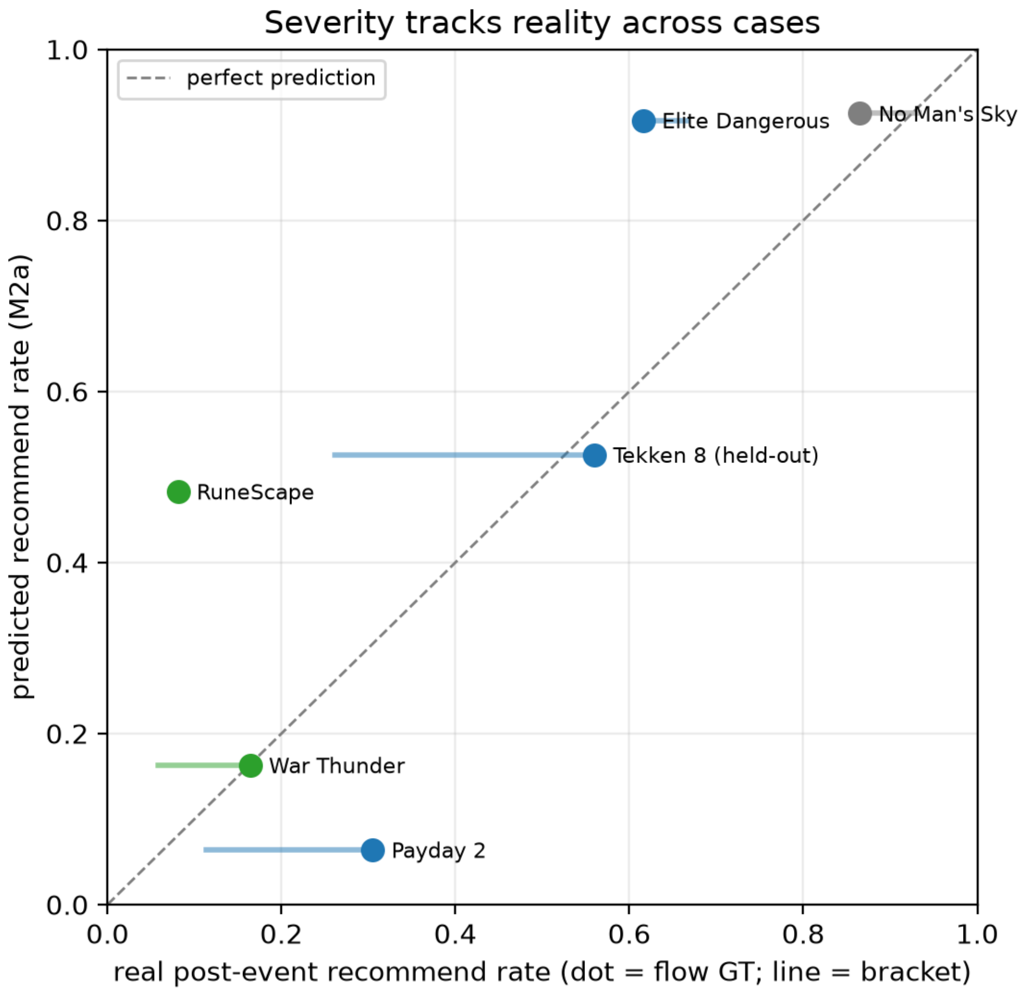
Elite Dangerous is the one case in the set where the bracket ends are reversed relative to the other four: the flow end (all reviews newly posted in the post-event window, n=211) is the *lower* number at 0.616, and the panel end (pre-event reviewers who edited their review in-window, n=24) is the *higher* number at 0.667. Panel is not always the lower bound: which end is which is a property of the case's own data, not a fixed convention, and is stated explicitly for that reason.

Umbrella tier: already-monetized game increases recurrent monetization

Case	Prior	Bracket [panel, flow]	M1	M2a	M2b	M3
War Thunder	0.772	[0.058, 0.164]	0.979 (JSD 0.5895)	**0.163** [0.116, 0.210] (JSD 0.0000)	0.171 (JSD 0.0001)	0.132 (JSD 0.0014)
RuneScape	0.861	[0.079, 0.081]	0.975 (JSD 0.7116)	**0.483** [0.396, 0.564] (JSD 0.1558)	0.460 (JSD 0.1417)	0.475 (JSD 0.1511)

RuneScape's bracket is unusually narrow (panel 0.079 vs flow 0.081) because the two structurally disjoint populations happened to agree almost exactly in this case: a near-unanimous revolt, not a population split by reviewing channel.

M2a lands inside or near its bracket on most cases, while M1 sits at the top regardless of the case.



Severity ordering tracks the diagonal.

Control tier: free content update, not monetization

Case	Prior	Bracket [panel, flow]	M1	M2a	M2b	M3
No Man's Sky	0.552	[0.933 panel, 0.865 flow]	0.973 [0.960, 0.984] (JSD 0.0304)	0.925 [0.871, 0.969] (JSD 0.0069)	0.932 [0.884, 0.970] (JSD 0.0089)	0.937 [0.907, 0.961] (JSD 0.0108)

Of the three controls run for this submission, only No Man's Sky is reported here as a scored result; CS:GO and Helldivers 2 remain in the evidence architecture (§3) as design controls but are not carried into this table. No Man's Sky's free "NEXT" update (2018-07-24) answers a question the five monetization cases cannot: is this pipeline only capable of tracking a *negative* swing? A method that always predicts backlash would fail this case outright. Instead, under the identical run configuration used throughout §5 (n=100, seed 42, pool of 3,000, permutation-averaged readout), the grounded panel predicted the redemption itself: approval rising from a 0.552 pre-event prior to a post-event bracket of [0.865 flow (n=8,645), 0.933 panel (n=2,394)], flow being the lower end here and panel the higher, the same ordering as Elite Dangerous's above and the opposite of the other four monetization cases: which end is which is a property of the case's own data, not a fixed convention. M2a lands at 0.925, inside the bracket and substantially closer to the outcome than to the prior. The order-position bias replicates here too, β median −0.47, the same sign as every scored case in this section.

Across all five scored cases, M1 (the ungrounded naive-prompting control) fails uniformly, sitting at 0.973–0.979 regardless of the case: it tracks neither the direction nor the magnitude of any real swing, which is exactly the floor it is designed to establish. The grounded methods (M2a/M2b/M3), by contrast, produce a predicted severity ranking that broadly tracks the real one: predicted M2a values of 0.064 (Payday 2), 0.526 (Tekken 8), 0.917 (Elite), 0.163 (War Thunder), and 0.483 (RuneScape) correspond to real flow ground truth of 0.305, 0.560, 0.616, 0.164, and 0.081 respectively. Two cases are essentially exact hits: War Thunder (0.163 vs 0.164; see the seed-robustness note in §5.6 on how much of this exactness is panel luck) and Tekken 8 (0.526 vs 0.560, the held-out case). One is a close under-shoot just below the panel bound (Payday 2), one is halved relative to the real collapse (RuneScape), and one reads as too rosy, predicting an improvement over the prior where the real reaction was a mild decline (Elite). The label-order bias identified in E1 (§5.6) replicates on every single case, consistently signed toward the later-listed answer option: median β of −2.44 (Payday 2), −3.53 (Tekken 8), −0.91 (Elite), −3.56 (War Thunder), and −4.25 (RuneScape).

5.2 The held-out case

Tekken 8's Tekken Shop cash-cosmetics store launched 2024-02-28, with a paid battle pass following in early April, an event that postdates Llama-3.3-70B-Instruct-Turbo's training cutoff (~December 2023) in its entirety. The model cannot have memorized the outcome; whatever the persona panel produces here is a genuine held-out prediction with no recall risk to caveat away. The backlash itself is a slow burn rather than a launch-day spike, which is why this is the one case scored on a 45-day window instead of 7. The window length was fixed from the shape of the real swing before any model was run on the case. Pre-event recommend rate was 0.835 (n=18,699); the post-event bracket is [0.261 panel (n=1,153), 0.560 flow (n=8,490)].

M2a lands at 0.526 [0.435, 0.619], essentially on the flow ground truth of 0.560, and the best result of the project. The raw option-logit gap has median −0.75 [−26.12, +19.12] with a standard deviation of 13.16: the model carries no strong normative slam on this case the way it does on Payday 2 or Elite, so the prediction is genuinely being carried by persona features rather than overridden by the model's own case-level opinion of the change. The disposition ordering (the coarse recommend-rate marker computed only from each persona's pre-event history) is the cleanest of any case scored and is the only one that crosses zero: personas with an all-recommend history score +4.28, most-recommend −11.31, half −13.58, and few (habitual critics) −20.11. The order-position bias replicates a third time here, β median −3.53. Where the model's own normative read of a case doesn't drown out the readout, as it does on Payday 2's promise-break or Elite's calm cosmetic add, grounded personas hit the real number almost exactly, with genuine population spread, on an event the model cannot have memorized.

5.3 Ordering / segment results

The central internal-consistency check is whether predicted answers order correctly along the disposition and exposure markers within a case even when the aggregate level is off. A method that gets the population's relative ordering right but the absolute level wrong is still doing something real, where one that gets neither right is not. That check holds up across every case scored in this submission. On Payday 2, the debiased judgment orders by monetization exposure as f2p-experienced −7.5 nats versus premium-only −14.0, and by disposition as all-recommend −10.9 versus habitual-critic (few) −21.3: f2p-experienced players and habitual optimists are both less severely affected than premium-only players and habitual critics, exactly as the construction hypothesis predicts. On Elite Dangerous the same disposition ordering holds with a wide separation: all-recommend personas sit at +21.85 versus few at +0.53. On the contrast case Total War 3, the separation is the widest observed anywhere in the project: few at −4.35 versus all at +19.39, a 24-nat gap, the marker's best showing of the project. Tekken 8's ordering (§5.2) is the cleanest of all, crossing zero cleanly at the all-recommend segment.

The reading this supports is the same one the negative results in §5.6 reinforce from the other direction: persona features order the population correctly and consistently, in every case tried, regardless of whether the model's overall verdict on that case's content lands close to, above, or below the real outcome. It is the case's own normative framing, not the persona construction, that sets the aggregate level.

5.4 Counterfactual decision menus

Two counterfactual menus re-survey the same validated personas against variant decision clauses instead of the real shipped one. Ground truth exists only for the shipped row in each menu; every other row is labelled extrapolation. The deliverable is the ranking and the differences between rows, not the absolute levels, since the absolute levels inherit the case-mood level problem documented in §5.6.

Payday 2 (shipped-row anchor 0.065, reproducing the ladder's 0.064):

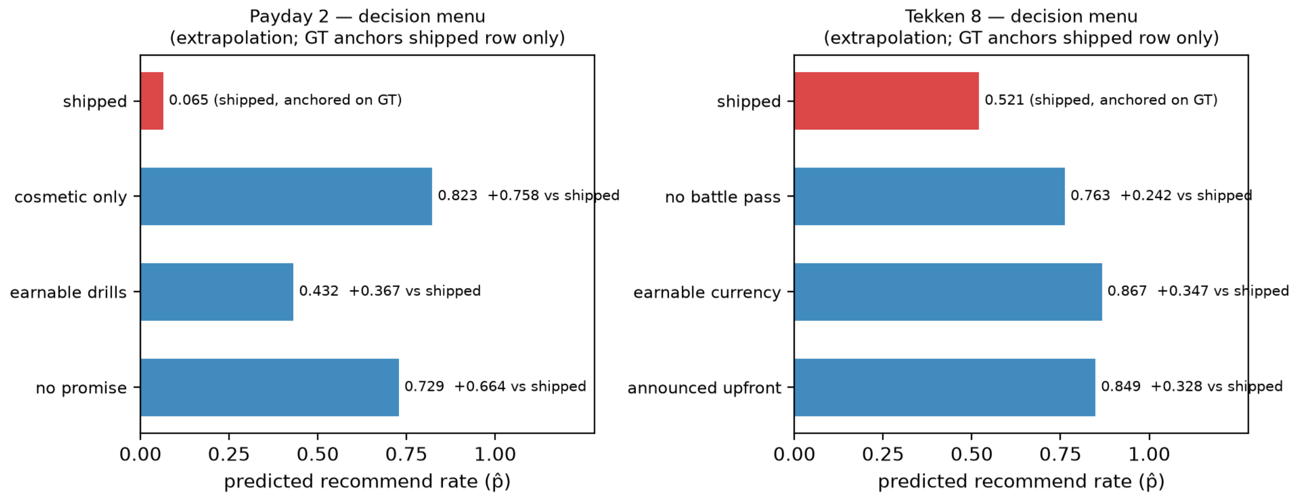
variant	$\hat{p}$	Δ vs shipped
shipped (broken promise + purchasable + stat bonuses)	0.065	-
cosmetic_only (no stat bonuses)	0.823	+0.758
no_promise (same MTX, no prior no-MTX promise)	0.729	+0.664
earnable_drills (stat bonuses, but not purchasable)	0.432	+0.367

Removing either the pay-to-win lever or the broken promise alone recovers most of the goodwill; the full collapse requires both together: a betrayal × pay-to-win interaction, not two independent penalties. The habitual-critic ("few") disposition segment stays at 0.00 in every variant, shipped or not.

Tekken 8 (shipped-row anchor 0.521, reproducing the ladder's 0.526, the more credible of the two menus, since its anchor sits essentially on ground truth):

variant	$\hat{p}$	Δ vs shipped
shipped (real-money shop + paid battle pass)	0.521	-
earnable_currency (shop currency earned via gameplay)	0.867	+0.347
announced_upfront (same shop + pass, disclosed before launch)	0.849	+0.328
no_battle_pass (shop only, no pass)	0.763	+0.242

Disclosing the shop and battle pass upfront buys back almost as much goodwill as removing the real-money lever entirely (+0.328 vs +0.347). The battle pass is the smaller of the two levers on its own: removing it and leaving the real-money shop in place (no_battle_pass) recovers +0.242, roughly a third less than the +0.347 recovered by making the shop's currency earnable, so the battle pass alone accounts for a meaningful but partial share of the total drop, with the real-money shop carrying the rest.



Every row but the shipped row is extrapolation beyond any ground truth.

5.5 The second question (words vs feet)

A second, independently scored question was put to the same individuals scored on the recommend question: "Would this player still be playing the game a month after this change?" The wording was frozen before the run. The ground truth is behavioural, not a review statistic. It is drawn from the scrape-time `author_last_played` field, used only as ground truth and never surfaced anywhere in the persona bio, over a fixed 30-day horizon. Because this is a real per-person rate rather than a structurally disjoint panel/flow split, there is no estimand bracket here; it is a single number checked against a single number, person by person.

The result is negative on both cases scored, and I'm reporting it as a central boundary rather than folding it quietly into the rest of the results. On Payday 2, the actual keep-playing rate was 0.850 (85/100 of the same personas); the model predicted 0.365, with person-level AUC of 0.278 (worse than chance) and accuracy at the 0.5 cut of 0.150. On Tekken 8, the actual rate was 0.740 (74/100); the model predicted 0.422, with AUC 0.514, indistinguishable from chance, and accuracy 0.270. The failure has a clear shape: on Payday 2 the model believes its most invested players are the ones most likely to quit, when in reality investment predicts staying: the "regular" investment segment is predicted to keep playing at only 0.22 but actually stays at 0.90. Rather than committing to a low probability, the model mostly hedges: 23/100 Payday 2 personas are predicted near zero and 70/100 land in the 0.4–0.5 decile; on Tekken 8, 79/100 land in that same hedging band.

The pattern across both cases is the review-bomb paradox made measurable: reviews collapsed into the bracket (0.113–0.305 on Payday 2, 0.261–0.560 on Tekken 8) while the large majority of the same people, 74 to 85 out of every 100, kept playing. The personas import the sentiment collapse wholesale into the behavioural prediction and cannot decouple what a vocal player says from what that same player does. The honest claim boundary this sets for the method: grounded personas forecast what vocal players say, not what they do. That makes the method usable for store-page and PR reaction forecasting, and not usable for churn or revenue forecasting.

5.6 Negative results & boundaries

This section belongs with the results, not tucked away as an appendix of caveats, because the failure modes below matter just as much to the method's honest claim as the hits in §5.1–5.4.

Label-position bias. Before the permutation-averaged readout was adopted, the 70B model favoured whichever answer option was listed second by a median of about 2.4 nats (range −8.75 to −0.25 across personas), measured directly in the E1 ablation. "Recommend" had sat first in every run up to that point, absorbing the penalty every time. Swapping to a Yes/No label pair did not fix it (the collapse persisted), showing this was a property of option *position*, not a property of the specific letter or word used as a label. The fix, the permutation-averaged readout itself, is mechanical and case-blind, decided before any ground truth was consulted for the fix, and was applied to every method including the M1 control so comparisons across the ladder remain fair.

The case-mood level problem. Even after that fix, the model's aggregate predicted *level* on a case is set largely by its own case-specific normative read of the change described, not by persona features alone. The sharpest evidence for this is a direct comparison between Payday 2 and Total War 3 (an out-of-scope paid-DLC contrast case), whose real outcomes are nearly identical (0.305 vs 0.321): on Total War 3 the model is confidently positive, M2a landing at 0.925 against bracket [0.116, 0.321] (a confident underreaction), while on Payday 2 it is confidently negative, overshooting the collapse. The debiased judgment gap tells the same story numerically: Total War 3's mean is +18.81 nats positive, Payday 2's is −13.50 nats negative, on cases the real world treated almost identically. Elite Dangerous shows the same problem from the over-forgiving side (§5.1): 0.917 reads as approval rising from the 0.721 prior, when the real reaction was a mild decline to 0.616–0.667.

Cross-case calibration does not transfer. A two-parameter sigmoid link fit on Total War 3's own ground truth ($\tau = 9.95$, $b = +24.28$), frozen, and applied to Payday 2's raw judgment gap gives 0.019 using the flow-end fit and 0.005 using the panel-end fit, nowhere near Payday 2's real bracket. A τ-sensitivity sweep in the transcript confirms no choice of τ rescues the transfer; the two cases' judgment-gap distributions sit on opposite sides of zero with different scales entirely (9.95 vs 3.92 standard deviation), which one shift-and-scale correction cannot reconcile. This is a direct consequence of the case-mood problem above: there is no single miscalibration to fix, because the thing that needs fixing is case-specific.

Frequency elicitation fails its own pre-registered test. Asking the model directly for a rate ("out of 100 players with this history, how many would recommend…") rather than reading the binary logprob gives 0.535 against Payday 2's bracket [0.113, 0.305], an overshoot from above, the opposite failure direction from the binary readout. 99 of 100 answers are multiples of 5, and 84 of 100 fall in {40, 60, 70}, heavy round-number clumping. The disposition ordering survives this elicitation method, but the exposure ordering is lost (premium 0.529 versus f2p 0.533, indistinguishable). This method is kept as a labelled contrast rung, not adopted as the scored readout.

The collapse is not a Llama-specific quirk. The same normative-confidence collapse that motivated the permutation-averaged fix shows up even more severely on DeepSeek-chat: the Recommend token falls below DeepSeek's top-20 logprob reporting floor for all 20 of 20 personas tested on Payday 2, a deeper collapse than anything observed on Llama, which at least kept the option within finite reporting depth.

The verbalized reasoning scaffold backfires. A logic-of-appropriateness scaffold (three fixed reasoning steps: "what kind of situation is this?", "what kind of person am I?", "what does a person like me do?", answered before the final response) was the strongest candidate surfaced by the prior-art scan, on the theory that reframing the question away from "is this good?" would sidestep the normative-collapse failure mode. The scaffold is drawn from the logic-of-appropriateness formulation in *Generative agent-based modeling with actions grounded in physical, social, or digital space using Concordia* (Vezhnevets, A.S., et al., Google DeepMind, arXiv:2312.03664). It made the level worse, not better: debiased judgment median −25.75 [−29.56, −10.75] versus plain M2a's −13.50, with $\hat{p}$ and spread both collapsing to 0.000. Auditing the stage-three reasoning shows why: the "what would a person like them do" answers re-derive the betrayal narrative directly from the situation description before ever reaching the final answer token, retriggering exactly the collapse the scaffold was meant to bypass. Not adopted.

The social graph (M3) adds nothing material. Across every case scored, M3 does not meaningfully outperform M2a or M2b. What it does provide is the homophily null-shuffle check, which is a genuinely informative diagnostic on its own: it holds (real spatial stance correlation exceeds the null) on Tekken 8 ($p=0.0050$), Elite Dangerous ($p=0.0050$), and RuneScape ($p=0.0050$), the three cases with the most genuine population-level stance variance to detect (M2a spread 0.471, 0.269, and 0.421 respectively), and is marginal or not supported on Payday 2 ($p=0.0647$, spread 0.151) and War Thunder ($p=0.1393$, spread 0.239), where the real stance distribution is more uniform.

Words versus feet. Cross-referenced in full in §5.5: the same personas that predict review sentiment accurately fail to predict real 30-day retention, with person-level AUC at or below chance on both cases tested. This is the method's clearest and most consequential scope boundary.

Panel-redraw robustness (three seeds). Every scored number above uses seed 42, fixed before any scoring run. To check whether those numbers depend on which 100 reviewers happened to be sampled, the identical frozen procedure (same pool, same construction, same permutation-averaged readout) was re-run twice more, under seeds 43 and 44, on all five monetization cases. M2a $\hat{p}$ [95% bootstrap CI] by seed (42 / 43 / 44):

Case	Seed 42	Seed 43	Seed 44
Payday 2	0.064 [0.034, 0.095]	0.029 [0.011, 0.051]	0.054 [0.029, 0.086]
Tekken 8	0.526 [0.435, 0.619]	0.513 [0.432, 0.591]	0.521 [0.434, 0.605]
Elite Dangerous	0.917 [0.858, 0.967]	0.933 [0.883, 0.974]	0.899 [0.843, 0.949]
War Thunder	0.163 [0.116, 0.210]	0.219 [0.170, 0.275]	0.217 [0.164, 0.269]
RuneScape	0.483 [0.396, 0.564]	0.470 [0.391, 0.551]	0.481 [0.394, 0.571]

Every qualitative verdict in §5.1 survives all three draws, and the label-order bias keeps its sign in all fifteen runs across the three seeds (median β from −0.91 to −4.44). Tekken 8 and RuneScape are the most stable of the five under redraw, moving by no more than about two points either way. The honest caveat is War Thunder: its third-decimal agreement with the flow ground truth at seed 42 (0.163 vs 0.164) is partly panel luck rather than a property of the method: the two redraws land at 0.217 and 0.219, both just above the bracket's upper end rather than on it, so the defensible statement is that War Thunder predicts at or just above the flow end (0.16–0.22 across the three draws), not that it hits it exactly. More generally, some redraws fall outside the seed-42 bootstrap CI entirely: Payday 2's seed 43 (0.029) sits below seed 42's [0.034, 0.095], and War Thunder's seed 43 and seed 44 both sit above seed 42's [0.116, 0.210], which means the bootstrap CI captures within-panel sampling noise (uncertainty from which 100 reviewers were drawn) but not panel-redraw variance (uncertainty from which *draw* of 100 reviewers was used), and should be read as a known understatement of the method's total uncertainty. Seed 42 remains the scored, reported run throughout this submission because it was fixed before any robustness run was made; seeds 43 and 44 are labelled robustness checks only and do not replace or adjust any number reported elsewhere.

6. Future work

The measured boundaries in §5.6 point toward a set of directions this submission did not build or run. I'm framing them here as directions, not promises.

- **The influence layer.** A critic/journalist voice layer on the M3 social graph was planned during the project and never built. Given that M3 as shipped adds nothing material over M2a/M2b (§5.6), whether an influence layer of this kind changes that conclusion is untested.

- **Injecting review-text signals without triggering persona collapse.** The current design withholds all review text from every bio, by construction, because it is the withheld verdict being scored against. A version of the method that could safely surface some review-text signal without leaking the target sentiment would need to guard against the failure mode documented in *The Chameleon's Limit: Persona Collapse & Homogenization in LLMs* (Xiao et al.): that distinct personas converge to a narrow mode, and that this collapse is axis- and domain-specific rather than uniform.
- **Untried readout candidates from the prior-art scan.** Answer-stem prefilling, PriDe-style label-free prior subtraction (per *Large Language Models Are Not Robust Multiple Choice Selectors*, Zheng, C., et al., arXiv:2309.03882), and behavioural-conditional question reformulation were all surfaced during the scan for alternatives to the permutation-averaged logprob readout but none was implemented or tested here.
- **A wider cross-model panel.** The cross-model check in this submission is a single DeepSeek probe (E3, §4.5, §5.6). Whether the case-mood level problem's severity varies systematically across a wider panel of models is open.
- **Scoring the no-swing tier, not just the swing tier.** The five scored monetization cases were deliberately chosen because they cleared a screening bar of a significant, measurable review swing: calibrating and validating the personas against a real outcome requires a real outcome to score against. That screen (§3, and documented case-by-case in the project's working notes) also surfaced several premium titles whose MTX additions produced no measurable approval swing at all (Rocket League, Rainbow Six Siege, Killing Floor 2, Dead by Daylight, and MGSV: The Phantom Pain among them), and those were dropped rather than scored. The consequence is that the validated case set, as it stands, conditions on a reaction having happened; it says nothing yet about how often a microtransaction addition produces no reaction at all, or how reliably the method recognizes that outcome. Deliberately adding that no-swing/mild tier as scored cases would do two things the current set cannot: measure the method's false-positive rate directly (does the panel predict calm when the true answer is "nobody measurably cared," beyond the two calm-direction checks already in the set: Elite Dangerous, No Man's Sky) and broaden the reference population from "cases that reacted" toward the unconditional distribution of MTX introductions generally. The latter is the necessary step before a persona population can honestly serve as a simulator for an untested product decision where no ground truth will ever exist.
- **Live pre-decision use.** The scored cases in this submission all run against the frozen historical dump; `llmsonas/data/ingest.py` is the live-API path that would let a studio run its own proposed change through the same ladder before shipping, as described in the FAQ's production-use answer (§8).

7. Communicating to the stakeholder

The deliverable's stakeholder-facing artifact is a decision brief, not a model card. That brief itself accompanies this submission as `stakeholder-brief.md`. Its structure is fixed regardless of which case populates it:

1. **Their question, restated.** Open with the stakeholder's own decision in their own terms ("you are considering adding [monetization change] to [game]") so the brief is read as an answer to a specific question, not a general research report.

2. **Precedents.** A short, named set of comparable real events (the pillar and umbrella cases) with their real, observed outcomes, establishing that this kind of decision has a measurable track record and is not unprecedented.

3. **The validated simulator.** Scored against real outcomes on five cases, the method produces two essentially exact hits (War Thunder 0.163 against a real 0.164; Tekken 8, held out by training-data

cutoff, 0.526 against a real 0.560), one close under-shoot just below the panel bound (Payday 2), one halved severity call (RuneScape), and one too-rosy call that reads as improvement where reality was a mild decline (Elite Dangerous). Naive, ungrounded prompting fails on all five, sitting at 0.97+ regardless of the case. The predicted severity ranking across the five cases tracks the real one. The honest bracket, not a single point estimate, is always the reported target.

4. **Lever ranking.** From the counterfactual decision menus, ranked by goodwill recovered: (1) keep new content cosmetic-only, avoid anything that affects competitive balance (+0.758 on Payday 2, the largest single lever moved); (2) never break a stated promise, or if disclosure is possible, make it upfront (+0.664 on Payday 2 for removing the broken promise; +0.328 on Tekken 8 for disclosing the shop and pass before launch, where upfront disclosure alone buys back almost as much goodwill as removing the real-money lever entirely, +0.347); (3) make purchasable content earnable through play instead of purchase-only (+0.367 Payday 2, +0.347 Tekken 8); (4) if a battle pass is being added on top of a shop, consider dropping that layer on its own: Tekken 8's shop-only variant (no battle pass, real-money shop retained) still recovers +0.242, a smaller but real share of the goodwill lost. These are extrapolated rankings, not scored predictions, and are labelled as such.
5. **Who flips.** The segments most likely to move from approve to disapprove are consistent across every case: players with a critical review history (the "few" or "half" disposition segments) are the most negative outcome in every case scored. On Tekken 8, the all-recommend disposition segment is the only one that stays net positive (+4.28), with every other segment negative. Habitual critics do not come back under any counterfactual either: Payday 2's "few" segment stays at 0.00 across every menu variant, shipped or not. Monetization-naive players (no prior exposure to free-to-play or mixed monetization) read a new MTX addition harder than players who already have f2p experience: on Payday 2, f2p-experienced personas sit at −7.5 nats versus premium-only personas at −14.0.
6. **Scope honesty.** An explicit statement of what the method can and cannot answer (it predicts vocal-reviewer sentiment, not revenue or churn: see §5.5 and the FAQ), stated as a limitation up front rather than discovered later.
7. **The offer.** What a follow-on engagement would look like: running the stakeholder's specific proposed change through the same validated ladder ahead of a real decision, rather than after the fact.

8. FAQ

How do you know the personas aren't just echoing the question's framing?

The survey question is one fixed, neutral, third-person sentence used identically across every case; the decision being evaluated lives entirely in the persona's situation clause, not in the question. The same fixed question produces opposite verdicts depending on which case's content is in the bio (a collapse on Payday 2, calm on Elite Dangerous, an exact middling call on Tekken 8), and the ungrounded M1 control, asked the exact same question with no persona grounding at all, sits at 0.97 or higher on every case regardless of content. That pattern is only possible if the answer is moving with the persona and the case facts, not with the question's wording. Where framing genuinely did leak into the result (the position of the answer options on the page), it was measured directly (E1, a median 2.4-nat bias) and removed with a mechanical, case-blind fix.

Isn't scoring against data the model may have memorized circular?

Personas are built exclusively from each reviewer's pre-event history, so the construction inputs cannot contain the outcome by design. The stronger answer is Tekken 8: its event (2024-02-28) postdates Llama-3.3-70B-Instruct-Turbo's training cutoff (~December 2023) entirely, so the model cannot have seen the outcome in training data at all, and it is also the best-scoring case in the set, 0.526 against a real 0.560. A result that strong on a case with zero memorization risk is hard to explain as recall.

What population do the personas actually represent (the estimand)?

Steam enforces one review per account, which makes the pre-event reviewer population that personas are built from and the in-window new-reviewer population that ground truth is measured from structurally disjoint sets of accounts. The nearest observable population to what the personas actually represent is the panel: existing pre-event reviewers who edited their review inside the post-event window. Both the panel end and the flow end (all reviews newly posted in-window) are reported as a bracket in every case, rather than picking one and presenting it as a single number.

Why logprobs instead of sampling answers?

Reading the model's option-token logprobs directly reads the belief itself rather than conflating it with sampling noise, and it is cheaper: one forward pass per persona per label order rather than repeated generations. The alternative was tested directly and failed: E2's frequency-elicitation variant, which asks the model to verbalize a rate instead, produces heavy round-number clumping (99 of 100 answers are multiples of 5) and overshoots the bracket from above rather than landing inside it.

Why is the target a bracket, not a number?

Because no single real population is simultaneously "who the personas are built from" and "who the post-event recommend rate is measured from": Steam's one-review-per-user rule makes those two populations structurally disjoint (§4.4). Reporting a single point estimate against either population alone would overstate the precision the underlying data actually supports; the bracket is the honest target, and every case in this submission reports both ends.

What happens on decisions the model has strong prior opinions about?

This is the method's main measured failure mode, not a hidden one: the case-mood level problem (§5.6). The model's own normative read of the change itself sets the aggregate level: confidently positive on Total War 3 (+18.8 nats) and confidently negative on Payday 2 (−13.5 nats debiased), on two cases whose real outcomes are nearly identical, and over-forgiving on Elite Dangerous. What survives even when the level is wrong is the ordering: disposition and exposure segments still rank correctly within every case. Permutation averaging removes the mechanical, position-based component of this problem; it does not and cannot remove the case-specific normative component, and no cross-case calibration link was found that does either (E4).

Can this predict revenue/churn?

No, and this was tested directly rather than assumed. The second scored question asked whether each persona would still be playing a month after the change, checked against real per-person behavioural ground truth. Person-level AUC came back at 0.278 on Payday 2 (worse than chance) and 0.514 on Tekken 8 (chance), even though 74 to 85 percent of the same people were still playing while their reviews collapsed. The method forecasts what vocal players say, which is validated; it should not be used for churn or revenue forecasting, which is refuted.

Why not fine-tune instead?

The claim this submission makes rests on held-out prediction: every scored number is produced from a model that never saw the outcome. Fine-tuning on post-event outcomes would turn that prediction into a fit and remove the falsifiability guard that makes the results defensible in the first place. The prior-art scan also considered survey-fine-tuning approaches and set them aside as unnecessary here: the observed failure mode is a case-level normative-mood problem, not a parameter-fitting problem, and E4 showed directly that a calibration link fit on one case's own ground truth does not transfer to another case, so there is no single miscalibration that more data or fine-tuning on it would obviously fix.

How much does a run cost?

A full scored case (the 70B model, n=100 personas, all four ladder methods, with the permutation-averaged readout doubling every survey call) runs to roughly $0.50 of API spend. The entire five-case scoring run reported in this submission, including six diagnostic ablations and both counterfactual decision menus and the behavioural second question, billed to roughly $1.76 in total.

What would production use look like (live API)?

The historical dump used for the scored cases in this submission exists to give breadth across many past events in one pass; it is not how a live deployment would work. `llmsonas/data/ingest.py` is the live-API ingestion path: it pulls reviews directly from Steam's public review API, and was the route used for the project's first end-to-end pass before the frozen dump became the primary data source for scored runs. A production engagement would use that path to pull a studio's current pre-event reviewer base directly from Steam, build the panel from it, and run the studio's actual proposed change through the same counterfactual-menu machinery (§5.4) before it ships, rather than reading the outcome off a review-bomb after the fact.

9. Repository map & reproduction

The codebase is a standalone Python package, `llmsonas/`, organized by pipeline stage:

- `llmsonas/config.py`: pinned run settings (model, target app, event cutoff, seed) for each case.
- `llmsonas/data/`: ingestion. `dump.py` reads the frozen historical Steam review dump used for the scored cases; `ingest.py` is the **live-API ingestion path**. It pulls reviews directly from Steam's public review API and was the route used for the project's first pass (the CS:GO smoke test) before the frozen dump became the primary data source for scored runs.
- `llmsonas/features/`: builds per-user behavioural feature vectors from the ingested data (library breadth, spend, tenure, review timing, etc.).
- `llmsonas/construction/`: persona construction: `select.py` (M2a/M2b sampling and archetype selection), `segment.py` (population-relative quantile segmentation), `exposure.py` (monetization-exposure marker), `profile.py` (facts-only third-person bio assembly).
- `llmsonas/survey/`: `prompt.py` builds the neutral third-person survey question and label layouts (including the swapped-order arm); `together_client.py` calls the hosted model and extracts option-token logprobs.
- `llmsonas/graph/`: the M3 homophily graph: `build.py` constructs the graph, `fj.py` implements the anchored Friedkin–Johnsen update, `contagion.py` and `nullcheck.py` support the contagion simulation and the null-shuffle significance check.
- `llmsonas/scoring/metrics.py`: Jensen–Shannon divergence, bootstrap confidence intervals, and swing error against the panel/flow bracket.

- `llmsonas/cases.py`: the parametrized per-case runner that wires ingestion, construction, the method ladder, and scoring together for a named case.
- `llmsonas/harness.py`: the shared execution harness (ladder orchestration, permutation-averaged readout).

`scripts/` holds entry points: per-case smoke tests (`smoke_test_*.py`), the diagnostic/ablation experiments (`e1_label_controls.py` through `e6_loa_scaffold.py`), the counterfactual decision-menu runner (`counterfactual_menu.py`), the second scored question (`q2_keep_playing.py`), and a reasoning-probe utility (`probe_reasoning.py`). `tests/` covers the readout, segmentation, and exposure-marker logic.

Running a case end-to-end, offline: with the frozen dump present, a case runs via its `scripts/smoke_test_<case>.py` entry point, which calls into `llmsonas/cases.py` to load the dump, build features, construct personas under M2a/M2b, run the M1/M2a/M2b/M3 ladder with the permutation-averaged readout, and print the scored table. No network access beyond the model API call is required once the dump is present locally.

Running live: `llmsonas/data/ingest.py` pulls fresh reviews directly from Steam's public review API rather than the frozen dump; this was the ingestion path used for the project's first pass and remains available for extending the case set to titles not covered by the historical dump.

A note on tooling: the implementation work in this project was largely done via agentic AI coding tools, with every design decision, scope call, and scored result directed and reviewed by me throughout.